

WEB STARS



LENYORA PRIMARY SCHOOL DIGITAL APPLICATION SYSTEM

Initial Software Requirements Specifications (SRS)

Submitted to:

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Department of Information Technology

In Partial Fulfilment of the Requirements for

Work Integrated Learning (WIL)

Submission Date: 26 August 2026

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Client Representative: Principal, Lenyora Primary School

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1. INTRODUCTION

1.1 Project Title

Lenyora Primary School Digital Application System

1.2 Problem Statement

Lenyora Primary School currently relies on fragmented manual processes to manage learner applications and communicate subject information to parents. School announcements are shared through physical notice boards and WhatsApp groups, subject listings are provided verbally or through printed handouts, and learner applications are collected and submitted using paper forms. This outdated approach creates significant challenges:

- **Communication Gaps:** Parents frequently miss important announcements due to the limitations of notice boards and the chaotic nature of WhatsApp group messages.
- **Lack of Transparency:** There is no centralized, accessible source for parents to view the subjects offered per grade, leading to confusion and misinformation.
- **Administrative Inefficiency:** School staff spend excessive time managing paperwork, manually tracking applications, and responding to repetitive parent inquiries.
- **Lost or Damaged Forms:** Paper application forms are often lost, misplaced, or damaged, causing frustration for parents and delays in the enrolment process.
- **No Digital Presence:** The school lacks a centralized digital platform where parents can reliably access information and complete applications online.

These challenges underscore the urgent need for a comprehensive digital solution that modernizes communication and application management at Lenyora Primary School.

1.3 Purpose, Aim and Objectives

Purpose

To develop a centralized digital system for Lenyora Primary School that digitizes all aspects of application announcements, subject information dissemination, and learner application submission and management.

Aim

To build a cross-platform Flutter mobile application for parents and a secure

ASP.NET Core web portal for school administrators, both integrated with a robust and scalable Supabase backend.

Objectives

To enable parents to view school announcements in real-time, including application opening and closing dates.

To allow parents to view accurate and up-to-date subject listings organized by grade.

To provide parents with an intuitive online form to apply for their children's enrolment.

To empower school administrators to efficiently manage announcements, subjects, and applications from a single web-based dashboard.

To ensure all system data is stored securely, reliably, and access-controlled using Supabase's PostgreSQL database and authentication services.

To improve the overall user experience for parents, especially those with limited digital literacy, through a simple and user-friendly interface design.

2. PROJECT SCOPE

2.1 Scope Definition

In Scope	
Component	Description
Mobile Application	A cross-platform Flutter application for parents/guardians, supporting Android 8.0 and above.

Admin Web Portal	An ASP.NET Core MVC/Web API accessible via modern web users.
Backend Infrastructure	Supabase for authentication, PostgreSQL database, and file storage.
User Authentication	Secure registration and login for both parents and school administrators.
Announcements Module	Admin CRUD operations; real-time viewing for parents on the app.
Subject Listing Module	Admin management of subjects per grade; parent viewing by grade.
Online Application Module	Parents can submit learner applications with required details and document uploads.
Application Status Tracking	Parents can check application status; admin can approve or reject applications.

Out of Scope	
Feature	Reason
Online School Fees Payment	Planned for a future phase; not required for initial deployment.
Learner Results Management	Academic records management is not part of the current phase.
SMS or Email Notifications	Notifications will be delivered through in-app announcements.
iOS App Publishing	The Flutter app will be built for Android only during this phase.

2.2 Deliverables and Acceptance Criteria

Deliverables

1. Initial SRS Document: This completed Software Requirements Specifications document.
2. Flutter Mobile Application: An installable Android APK file that can be distributed to parents.

ASP.NET

3. Admin Website: A fully functional, deployable web application

for school administrators.

4. Supabase Database Setup: A configured Supabase project with defined tables, relationships, security policies (Row Level Security), and storage buckets.

Acceptance Criteria

#	Criterion
AC1	Parents must be able to register, log in, and successfully submit a learner application with all required fields and document uploads.
AC2	When an admin creates or updates an announcement, it must appear on the parent app within 2 seconds (real-time sync).
AC3	An admin must be able to add, edit, and delete subjects and assign them to specific grades.
AC4	All submitted applications must be stored in Supabase and be immediately viewable, searchable, and manageable by the admin on the web portal.

2.3 Exclusions, Constraints and Assumptions

Exclusions

- No payment gateway integration.
- No SMS or email notification services.
- No iOS app publishing (Android only).
- No integration with third-party school management systems.

Constraints

- Technology Stack: The project must be built using Flutter for the mobile app, Core **ASP.NET** (MVC/Web API) for the web portal, and Supabase as the backend.
- Platform Support: The mobile app must run on Android 8.0+. The admin portal must be compatible with the latest Chrome and Edge browsers.
- Timeline: The initial SRS must be submitted by 25 August 2026; project delivery expected by the end of the semester. Assumptions
- Parents/guardians have access to smartphones with a stable internet connection.
- The school has a dedicated PC with a reliable internet connection for administrative use.
- The stakeholder (Principal/Admin) will provide all official school information (grade structure, subject lists, application dates) within the stipulated time.
- The school will designate a staff member to be the primary system administrator.
- Users have basic literacy in English to navigate the application.

3. STAKEHOLDERS AND USER GROUPS

Role	Description	Key Responsibilities/Interests
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Primary Stakeholder	Principal / Administrator of Lenyora Primary School	Overall approval, providing school data, and ensuring the system meets school needs.
Parent/Guardian	Primary end-user of the Flutter app	View announcements and subjects; apply for child; track application status.
School Admin	School staff managing the system	Manage announcements; manage subjects; review, approve/reject applications.
Guest	Unauthenticated website visitor	View general school information (name, contact, grade list) without login.
Development Team	Students and project supervisor	Design, develop, test, and deploy the system.

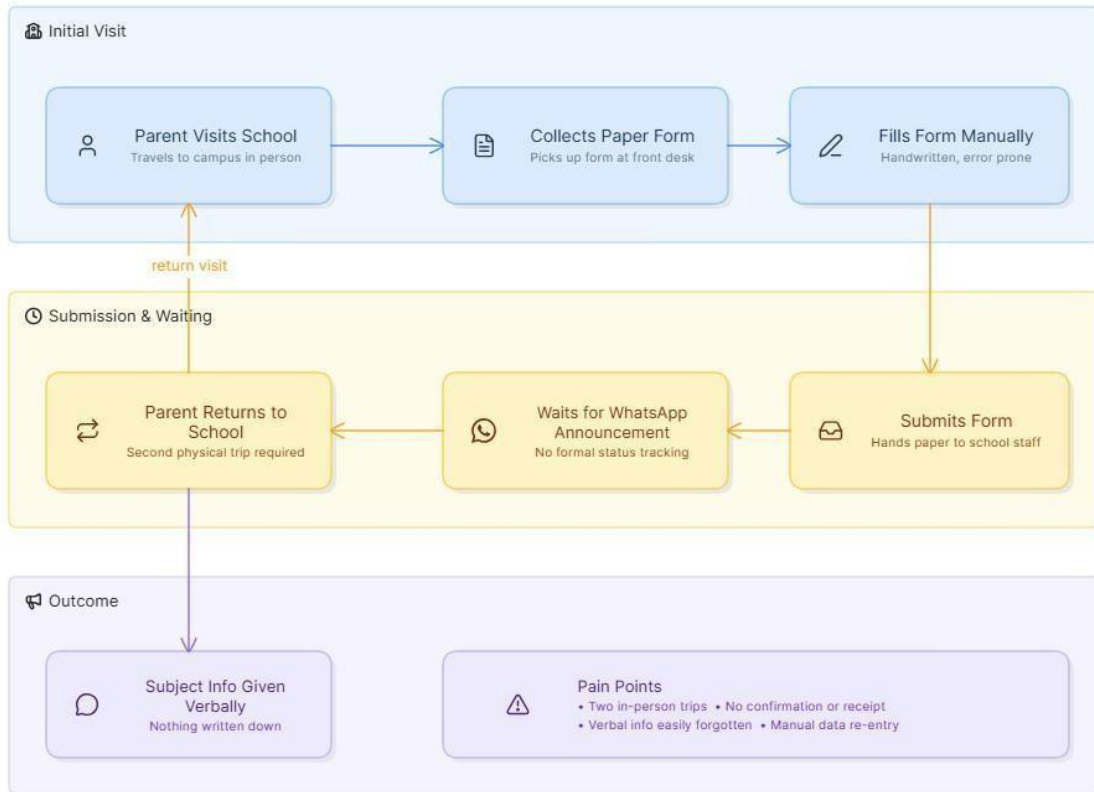
4. CURRENT PROCESS AND PROPOSED PROCESS

4.1 Current Process (As-Is)

The following flowchart illustrates the current manual process:

Manual School Enrollment Process

Current paper-based parent workflow



Detailed Steps:

Parent visits the school campus to inquire about admissions.

Parent collects a paper application form from the administrative office.

Parent manually fills out the form and gathers required physical documents (ID, birth certificate).

Parent returns to the school to submit the completed form and documents.

Staff processes the paperwork manually.

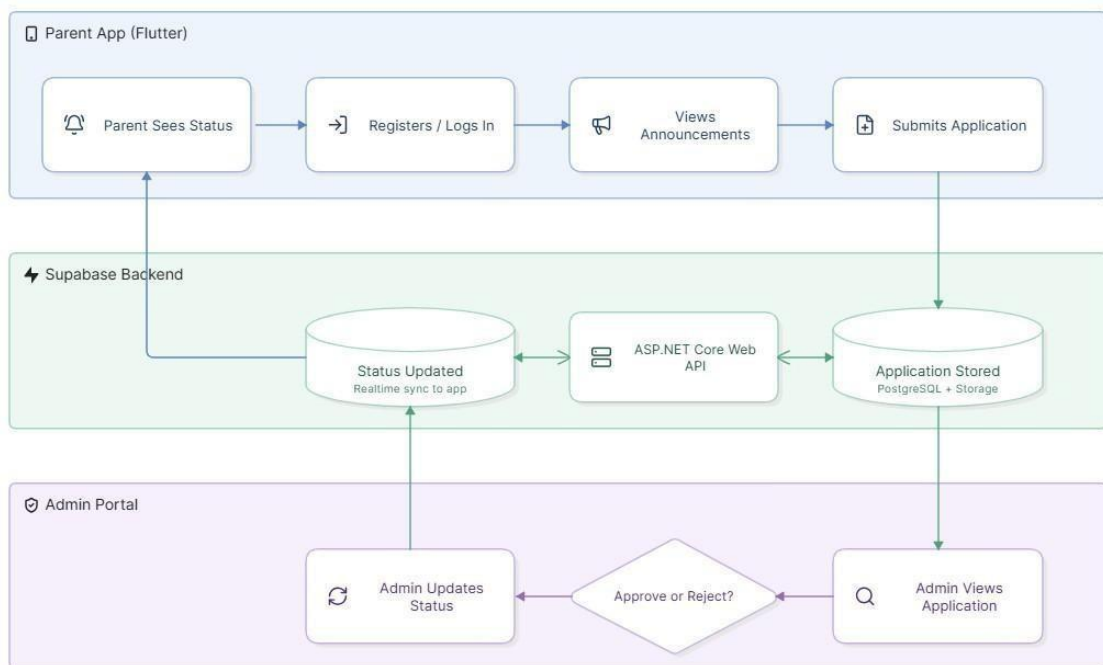
Parent waits for a WhatsApp message or notice board announcement regarding the application outcome.

Subject information is provided verbally or through printed handouts.

4.2 Proposed Process (To-Be)

The following flowchart illustrates the proposed digital process:

Parent Enrollment Flow



Detailed Steps:

Parent opens the Lenyora Primary School Flutter application.

Parent registers and logs into the application.

Parent views current announcements, including application opening and closing dates.

Parent browses subject listings per grade.

Parent fills out the digital application form and uploads required documents (e.g., ID, birth certificate).

Parent submits the application; data is stored in Supabase.

Admin is notified of the new application on the **ASP.NET** web portal.

Admin reviews the application details and documents.

Admin approves or rejects the application, updating its status.

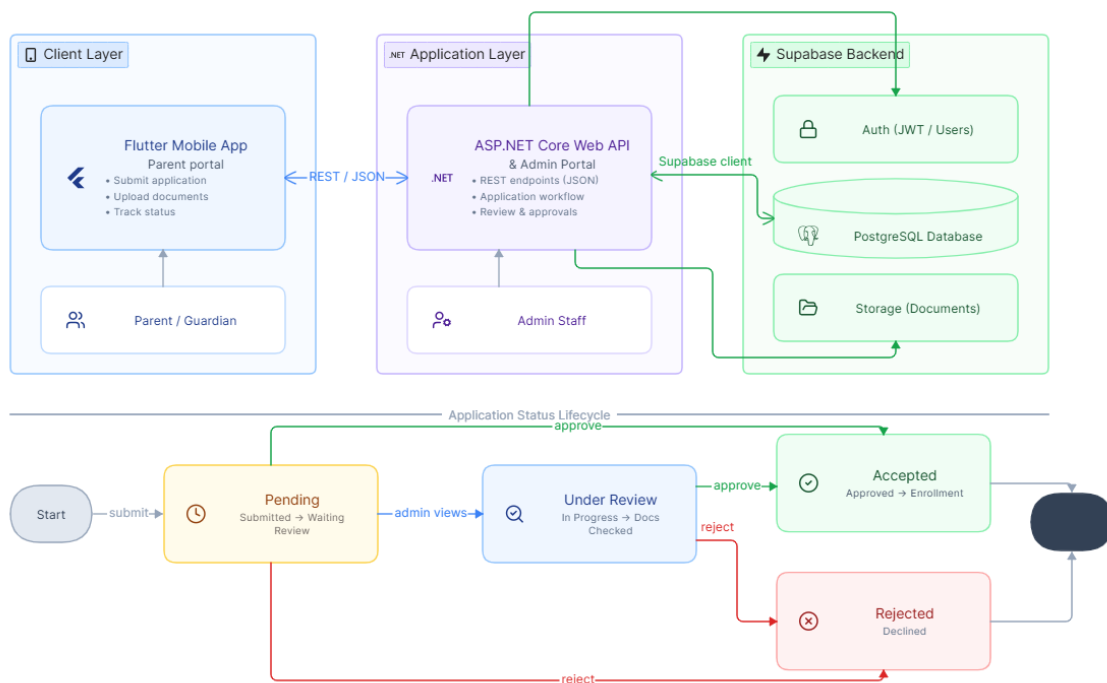
Parent opens the app, navigates to the status tracker, and sees the updated status (Pending/Approved/Rejected).

5. PRODUCT PERSPECTIVE

This system is a new, standalone solution designed to replace the school's current manual processes. It is a greenfield project with a modern, cloud-based architecture to ensure scalability, security, and maintainability.

System Architecture Diagram

School Enrollment System — Architecture



Key Layers

Presentation Layer:

Flutter App: Provides the user interface for parents, offering a clean and intuitive mobile experience.

ASP.NET Admin Portal: Provides a responsive web interface for school administrators to manage the system.

Business Logic Layer:

ASP.NET Web API: Handles core business logic, data validation, and API endpoints consumed by the Flutter app.

Data Layer:

Supabase: Acts as the primary data store, providing authentication, PostgreSQL database, and file storage with robust security features.

6. MAIN FLUTTER, **ASP.NET** AND SUPABASE FUNCTIONS

6.1 Flutter (Mobile Application)

Function	Description
<i>PARENT REGISTRATION/LOGIN</i>	Secure sign-up and sign-in using Supabase Auth.
<i>ANNOUNCEMENTS</i>	Display a list of announcements with real-time updates. Show opening and closing View dates.
<i>VIEW SUBJECT LISTINGS</i>	Display subjects organized by grade.
<i>APPLY FOR CHILD</i>	Provide a multi-step form for entering child details (Name Age, Grade) and parent details .Allow document uploads (ID, birth certificate).
<i>TRACK APPLICATION STATUS</i>	Display status of submitted applications (Pending/Approved/Rejected) in a user friendly format.

6.2 **ASP.NET** Core (Web API & Admin Portal)

FUNCTION	DESCRIPTION
ADMIN AUTHENTICATION	Secure login for authorized school administrators.
MANAGE SUBJECTS (CRUD)	Add, edit, delete, and assign subjects to grades.
MANAGE ANNOUNCEMENTS (CRUD)	Create, update, delete, and schedule announcements.

VIEW APPLICATIONS	Display all applications with detailed information.
APPROVE/REJECT APPLICATIONS	Change application status with an option to add notes/comments.
ADMIN DASHBOARD	Display key metrics and statistics (e.g., total applications, pending approvals).
API ENDPOINTS	Expose restful APIs for the Flutter app to consume.

6.3 Supabase (Backend)

Function	Description
Authentication	Manage user sessions and JWT tokens for parents and admins.
PostgreSQL Database	Store and manage data in tables: user , s announcements, subject , application . s s
Row Level Security (RLS)	Enforce data access policies to ensure users only see their own data.
Storage Buckets	Store uploaded documents (ID, birth certificates) securely.
Real-time Subscriptions	Enable instant updates of announcements on the Flutter app.

7. FUNCTIONAL REQUIREMENTS

ID	Requirement	Priority
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FR1	The system shall allow parents/guardians to register and log in securely using email and password.	Must Have
FR2	The system shall allow an admin to create, update, and delete announcements.	Must Have
FR3	The system shall display announcements on the parent app in real-time (within 2 seconds).	Must Have
FR4	The system shall allow an admin to manage subject listings (add, edit, delete) per grade.	Must Have
FR5	The system shall allow parents to view subject listings filtered by grade.	Must Have
FR6	The system shall allow a parent to submit a learner application with required details (Child Name, Age, Grade, Parent Details) and upload documents.	Must Have
FR7	The system shall allow a parent to view the status of their submitted applications (Pending/Accepted/Rejected).	Should Have
FR8	The system shall allow an admin to view, search, filter, and approve/reject applications.	Must Have
FR9	The system shall allow an admin to search and filter applications by status, grade, and applicant name.	Should Have
FR10	The system shall validate required form fields before submission.	Must Have
FR11	The system shall display appropriate error messages for failed operations (e.g., login failure).	Must Have

8. NON-FUNCTIONAL REQUIREMENTS

ID	Requirement	Detail
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NFR1(Performance)	The app and website should load in < 3 seconds on a standard internet connection.	High priority for user satisfaction
NFR2(Security)	Passwords must be encrypted (hashed). Supabase Row Level Security must be enforced. Only admins can delete data.	Critical for data protection.
NFR3(Usability)	The interface must be simple and intuitive, usable by parents with low digital literacy. Clear labels and guided workflows.	Essential for adoption
NFR4(Availability)	System should be available 99% of the time during school hours (8 AM – 4 PM)	Critical for continuous service
NFR5(Scalability)	System should handle up to 500 concurrent parent users during peak application season	Ensures smooth performance
NFR6(Maintainability)	Code should be well-documented and follow standard coding conventions for easy maintenance.	Important for future development
NFR7(Mobile Responsive)	The admin portal should be responsive and accessible on different screen sizes.	For flexibility of admin access

9. SYSTEM REQUIREMENTS

9.1 Software Requirements

Component	Technology
Mobile App	Flutter 3.22+ (or latest stable version)
Admin Portal/ API Component	.NET 8 ASP.NET Core MVC and Web API)
Backend	Supabase (latest stable version)

Mobile OS	Android 8.0 (API level 26) and above
Admin Web Browser	Latest Google Chrome / Microsoft Edge

9.2 Hardware Requirements

User	Minimum Specification
Parent (Mobile)	Android smartphone with 2GB RAM, 16GB storage, and a camera (for document scanning)
School Admin	PC with 4GB RAM, 1GHz processor and a 10Mbps internet connection

9.3 Network Requirements

- A stable internet connection is required for both the mobile app and the web portal.
- The mobile app should handle offline scenarios gracefully (e.g., show cached data and notify user of network loss).

10. INTEGRATION REQUIREMENTS

Integration Point	Description
Flutter App ↔ ASP.NET Web API	The Flutter mobile app will communicate with the backend through REST APIs (JSON format) hosted on the ASP.NET Web API.

ASP.NET API ↔ Supabase	The <u>ASP.NET API</u> will use the official Supabase Client library for .NET to interact with the database, authentication, and storage.
Flutter App ↔ Supabase	The Flutter app will use the Supabase Flutter client for direct authentication (login/register) and real-time subscriptions for announcements
Flutter App ↔ Supabase Storage	The Flutter app will upload application documents directly to the Supabase Storage bucket.
Authentication Flow (PC)	Supabase Auth will manage authentication. The Flutter app and ASP.NET _API validate requests using Supabase JWT tokens.

11. REQUIREMENTS PRIORITISATION (MoSCoW)

Priority	Requirement
Must Have	Parent Registration/Login; View Announcements per Grade; Apply for the child; Admin Manage Applications; Admin Manage Announcements
Should Have	Documents Upload for Applications; Applications status tracking; Search and filter Applications (Admin)

Could Have	Push Notifications for new announcements; Download Applications as PDF; Email notifications for status Updates
Won't Have	Online Fees Payment; Chat with the Teacher functionality; iOS App publishing

12. APPENDICES

Appendix A: Letter of Agreement with Lenyora

Primary School Principal

17 August 2026

WIL PROGRAMME- WEB STARTS

TO WHOM IT MAY CONCERN

Lenyora Primary School agrees to be the stakeholder for Web Stars group from Central University of Technology for the IT Work Integrated Learning program.

Our company is willing to work with the group to achieve good quality output through the development of the School Digital Application System for Lenyora Primary School.

Regards,



Head of Department

Mr TP Moahloli



Appendix B: Stakeholder Communication Evidence

Meeting Summary

Date: 24 August 2026

Attendees: Project Team, Principal of Lenyora Primary School
Location: Lenyora Primary School, Administration Office Key Points
Discussed:

- Confirmed the need for a mobile application for parents and a web portal for the school.
- Discussed current challenges with paper-based applications and communication.
- Agreed on key system features (announcements, subjects, applications).
- Identified primary users and their needs.
- Principal provided initial school data (grade structure, subject lists).
- Signed a letter of agreement for the project.



Appendix C: Interview Notes

Interview with: Principal, Lenyora Primary School

Date: 24 August 2026

Key Requirements Identified:

1. "We need a way to communicate with parents instantly and reliably."
2. "Parents often ask which subjects are offered in each grade.
We need to make this easily accessible."
3. "The application process is too manual. We lose forms and parents lose forms. We need a digital solution."
4. "It would be helpful to see all applications in one place and easily track them."
5. "We want to be able to update application status and have parents see it immediately."
6. "The app should be simple to use. Many parents are not very familiar with technology."
7. "We need the system to be affordable and easy to maintain."

Appendix D: GitHub Repository

GitHub Repository:

<https://github.com/hlonikayane/SCHOOL-DIGITAL-APPLICATION->

[SYSTEM.git](#)

Appendix E: Supabase Database Screenshot

School Digital App...FREE

School Digital App...mainPRODUCTION

Connect

Feedback

Search

Home

Dashboard

Database

API

Auth

Storage

Settings

School Digital Application System

<https://tociqhbmzenslitskaau.supabase.co> Copy

STATUS

Healthy

COMPUTE

NANO

GITHUB

No repository connected

RECENT BRANCH

No branches

LAST MIGRATION

No migrations

LAST BACKUP

No backups

Primary Database

West EU (Ireland)

eu-west-1 · t3.nano

CPU 2% · Disk 13% · RAM 48% · 7/60 conns

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Term	Definition
SRS	Software Requirements Specifications
Flutter	Google's open-source UI toolkit for building natively compiled applications
API (Application Programming Interface)	A set of rules and protocols for building and interacting with the software applications
REST (Representational State Transfer)	An architectural style for designing networked applications, relying on stateless, client-sever communication, typically over HTTP.
JWT (JSON Web Token)	A compact, URL-safe token used for securely transmitting information between parties
CRUD (Create, Read, Update, Delete)	The four basic operations for persistent storage
Row Level Security (RLS)	A database security feature that restricts access to rows in a table based on the user executing a query
MoSCoW	A prioritisation technique: Must Have, Should Have, Could Have, Won't Have across mobile, web and desktop from a single codebase.
ASP.NET	A cross-platfrom; high-performance framework by Microsoft for building Core modern, cloud-based, internet-connected applications

Supabase	An open-source Firebase alternative that provides a PostgreSQL database, supabase authentication, storage and real-time capabilities.
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REVIEWING CURRENT PROJECT

Question	Group response
What is the main problem being addressed?	Lenyora Primary School's manual, paperbased processes for learner applications and communication are inefficient, leading to lost forms, communication gaps, and administrative burden.
What are the main Flutter functions?	Parent registration/login, viewing announcements, submitting learner applications, and tracking application status.
What are the main ASP.NET functions?	Admin authentication, managing announcements (CRUD), viewing applications, and approving/rejecting applications.
What information will Supabase manage?	User authentication data, announcements, applications, and uploaded documents (e.g., IDs, birth certificates).
Which functions appear most difficult?	Real-time announcement updates, secure document uploads, and building a simple, intuitive UI for users with low digital literacy.
Which functions require skills the group still needs to develop?	Skills in Flutter, ASP.NET Core, and being Supabase are c learned with the project.
Which functions may require more time than is available?	Building a fully polished, well oiled productionready mobile app and a website with all 'Must Have' and 'Should Have' features within the semester.

TECHNICAL FEASIBILITY

Question	Group finding
Does the group have the required Flutter skills?	Yes
Does the group have the required ASP.NET skills?	Yes
Can the group work with Supabase?	Yes
Can Flutter and ASP.NET connect to the same Supabase backend?	Yes
Are suitable development computers available?	Yes, each member has a personal computer suitable for the development
Are suitable test devices available?	Yes, the group members have android smart phones for testing
Is reliable internet access available?	Yes, all members have reliable internet access at their rooms and in campus
Which technical skills still need to be learned?	Practical skills in Flutter (UI, state management and API) ASP.NET (MVC, API, Integration Supabase), Supabase (Database design)
Which functions may be too technically difficult?	Implementing secure documents may be challenging

Technical Feasibility Decision:

Feasible with changes.

Explanation for the decision:

We are going to allocate significant time to finish the project.

OPERATIONAL FEASIBILITY

Question	Group finding
Will the prototype address the stakeholder's problem?	Yes
Who will use the mobile application?	Learner's parents
Who will use the web application?	School's admin staff
Do the intended users have suitable devices?	Most
Do users have suitable internet access?	Majority

Will the proposed solution fit the organisation's current activities?	Yes, it's going to help them save more time
Will users require guidance or training?	No, there will be instructions for first time users so the system will be easy to use
Are any proposed functions unsuitable for the intended users?	No, the system will be simplified for the intended users

Operational Feasibility Decision:

Feasible

Explanation for the decision:

The system is designed to solve real stakeholder problems and will fit into the school's current activities.

ECONOMIC FEASIBILITY

Resource	Available?	Action required
Internet access or data	Yes	Making sure we have sufficient data at home and campus
Transport to the stakeholder	Yes	We book necessary transportation for our meetings with the stakeholder
Development computers	Yes	We installed the required software to ensure the devices are ready for use
Test devices	Yes	Charge devices and update the operating system
Required software	Yes	Update to the latest stable versions to avoid compatibility issues
Supabase resources	Yes	Configure project limits

Presentation resources	Yes	Create PowerPoint slides, ensure projector works and have backups in case of internet failure
Other project resources	No	N/A

Economic Feasibility Decision:

Feasible.

Explanation for the decision:

We are currently in position with all necessary resources; software development hardware and internet.

SCHEDULE FEASIBILITY

Question	Group finding
Is enough time available for system design?	No
Is enough time available for Flutter development?	Yes
Is enough time available for ASP.NET development?	Yes
Is enough time available for Supabase integration?	Yes
Is enough time available for testing and corrections?	Yes
Is enough time available for stakeholder feedback?	Yes. We are going to schedule followup meetings
Are too many functions currently included?	No. We removed the lower priority functions

Which functions may need to be simplified, postponed or removed?	We simplified the function of uploading documents and removed the function of appointment booking
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Schedule Feasibility Decision:

Feasible with changes.

Explanation for the decision:

Some functions had to be removed.

LEGAL FEASIBILITY

Question	Group finding	<u>Legal Feasibility Decision:</u>
Will the prototype collect personal information?	Yes	
Is all the information necessary?	Yes. All information will be necessary and useful.	
Who should be allowed to access the information?	Only the admin staff responsible for reviewing the applications	
Will test data be used during development?	Yes	
How will passwords and private keys be protected?	We are going to use cryptographic hashing functionality	
Does the project involve confidential stakeholder information?	No	
Is private stakeholder information being kept out of GitHub?	Yes	

Feasible.

Explanation for the decision:

We are aware of the data privacy requirements and we are going to demonstrate appropriate security considerations.

ETHICAL FEASIBILITY

Question	Group finding
Does the stakeholder understand the purpose of the project?	Yes. We interviewed the stakeholder and he gave us the approval letter.
Is unnecessary personal information being collected?	No
Could the proposed solution negatively affect intended users?	No. It will have a positive impact on the intended users instead.
Is information being used only for its intended purpose?	Yes
Does the proposed prototype accurately reflect stakeholder needs?	Yes
Are users being treated fairly and responsibly?	Yes, they will be treated fairly and responsibly

Ethical Feasibility Decision:

Feasible.

Explanation for the decision:

We aim to ensure that the system serves the needs of its users without causing any harm.

FEASIBILITY SUMMARY

Feasibility Area	Decision	Main Reason	Change Required
Technical	Feasible with changes	We are going to allocate significant time to finish the project	Allocate more working hours dedicated to the project
Operational	Feasible	The system is designed to solve real stakeholder problems and will fit into the school's current activities	None
Economic	Feasible	We are currently in position of all necessary resources; software development hardware and internet	None
Schedule	Feasible with changes	Some functions had to be removed	We'll be prioritizing essential functionalities within our limited timeframe

Legal	Feasible	We are aware of the data privacy requirements and we are going to	None
		demonstrate appropriate security considerations	
Ethical	Feasible	We aim to ensure that the system serves the needs of its users without causing any harm	None

OVERALL FEASIBILITY DECISION

The proposed project is feasible.

The main strengths of the project are that we are aligned with the stakeholder's needs, we have a well-defined problem statement and the technology to solve the problem.

The main limitations identified are mastering the technology within a limited timeframe.

To ensure that the project can be completed within the semester, the group will prioritize the development of the essential 'Must Have' functions (like user auth, application submission for parents, application management for admins, such as document uploads, and real-time status tracking, until the core system is stable and functional.

Project Risk Register

Picture1 Picture2

R6	Internet connectivity may be unreliable for team members	Resource	1	3	3	Low	Ensure all team members have backup plans(e.g. use campus Wi-Fi or Mobile Data)	Use a mobile hotspot as a backup for our PC	L Kayane
R7	Data privacy or security oversight could expose personal information	Data & Security	2	3	6	Medium	Implement and strictly enforce Supabase Row Level Security policies	Notify the stakeholder immediately and restrict data access until the issue is resolved	MN Matla
R8	Inadequate testing may lead to a prototype with unresolved bugs	Quality	2	2	4	Medium	Plan for a dedicated testing period in the Microsoft Project Schedule	Postpone low priority features to create more time for testing bug fixes	GD Sekhula
R9	Users may struggle with lower digital literacy	Operational	2	2	4	Medium	Design the mobile app with a very simple guided user interface	Create a short, easy-to-understand user guide for the app	L Kayane
R10	Compatibility issues with different Android versions	Technical	2	2	4	Medium	Test the app on a range of Android emulators and physical devices covering the range API level (Android 8.0+)	Provide support for the most common Android versions first	MN Matla
R4	A group member may not complete assigned tasks on time	Team	2	3	6	Medium	Divide tasks and specific tasks to each member with defined deadlines	Reassign urgent tasks to another member and record the issue in the project progress tracker	SM Nkosi
R5	Development tasks may take longer than planned	Schedule	2	3	6	Medium	Create a detailed schedule with a realistic time estimate for each task	Prioritize "Must Have" features and be ready to postpone "Should Have" features	All Members

Highest Risks

Highest risk	Why is this important?	What must the group do now?
1. R3 - Too many proposed functions	If the team over-promises on features, they may end up with an unfinished product that doesn't meet the core 'Must Have' requirements.	The group must formally review the functional requirements and strictly prioritize 'Must Have' functions.
2. R1 - Limited Supabase experience	Supabase is the backbone of the system.	The group should immediately prioritize a "spike" or proof-of-concept to set up a basic Supabase project and connect both <u>ASP.NET</u> and Flutter applications to prove the concept works.
3. R4 - Group member may not complete assigned tasks	Threatens the project timeline.	The Project Lead should establish a clear and regular progress tracking system with brief daily stand-up meetings to monitor contributions and identify blockers early.

SWOT ANALYSIS: Lenyora Primary School Digital Application System

INTERNAL FACTORS

STRENGTHS	WEAKNESSES
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Clear Stakeholder engagement: The group has secured a real external stakeholder (Lenyora Primary School) with a signed agreement and documented interview confirming the project's need. Well-defined requirements: The SRS documents clear functional and nonfunctional requirements based on stakeholder interviews. Modern technology stack: Flutter, ASP.NET Core, and Supabase are appropriate and wellsuited to the project requirements. Good group collaboration: The Project Progress Tracker shows all members contributed during Phase 1, with clear task division and resolved challenges. Appropriate resources: All necessary development tools and test devices are available at no cost.	Tight schedule: One semester is available, but the group must learn new technologies while developing the prototype. Limited project management experience: The group is still learning tools like Microsoft Project, which may affect planning accuracy.
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EXTERNAL FACTORS

OPPORTUNITIES	THREATS
Real stakeholder needs: The school faces genuine problems with paperbased processes and is eager for a digital solution. Growing market for school digital solutions: Similar systems are increasingly adopted by schools, indicating demand. Skill development: This project provides valuable practical experience with in-demand technologies.	Scope creep: The stakeholder may request additional features during development, threatening the schedule. Technology changes: Updates to Flutter, ASP.NET Core, or Supabase could require additional learning or affect compatibility. Infrastructure limitations: Parents may lack suitable Android devices or reliable internet, affecting adoption.

PHASE 2 — SYSTEM ARCHITECTURE AND DESIGN

1. System Architecture Design

The proposed system is a three-part integrated solution. The Flutter mobile application serves parents/guardians; the ASP.NET Core web application serves authorised school administrators; and Supabase provides the shared backend services. Supabase Authentication handles registration, login, sessions and JWT authentication; PostgreSQL stores shared system data; Storage holds uploaded application documents; and Realtime supports announcement/status updates.

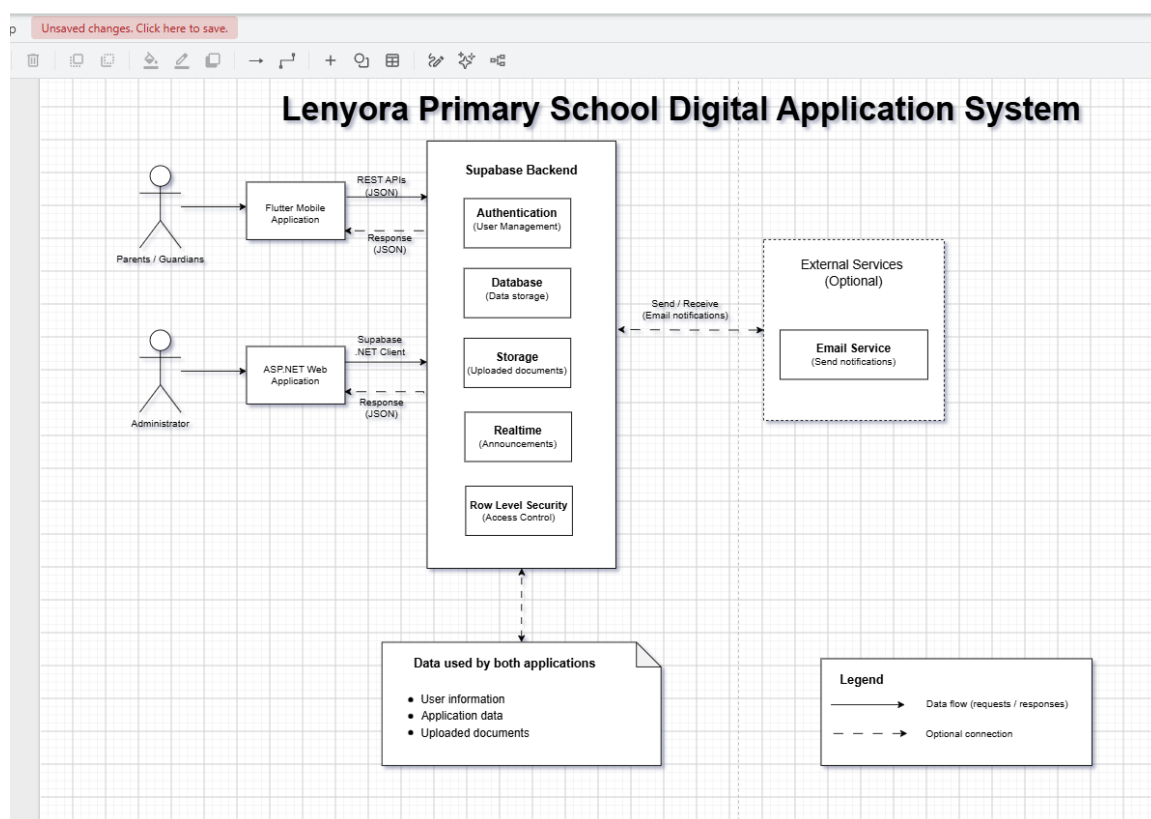


Figure 1. System architecture and major data flows.

2. Component Responsibilities

Component	Responsibility
Flutter Mobile App	Parent/guardian registration and login; announcements; subjects by grade; learner application; document upload; application-status tracking.
ASP.NET Core Web App / Web API	Administrator login; dashboard; announcement/subject CRUD; application viewing/search/filter/review; approve/reject; notes; REST API operations.
Supabase Auth	Registration, login, sessions and JWT-based authentication.

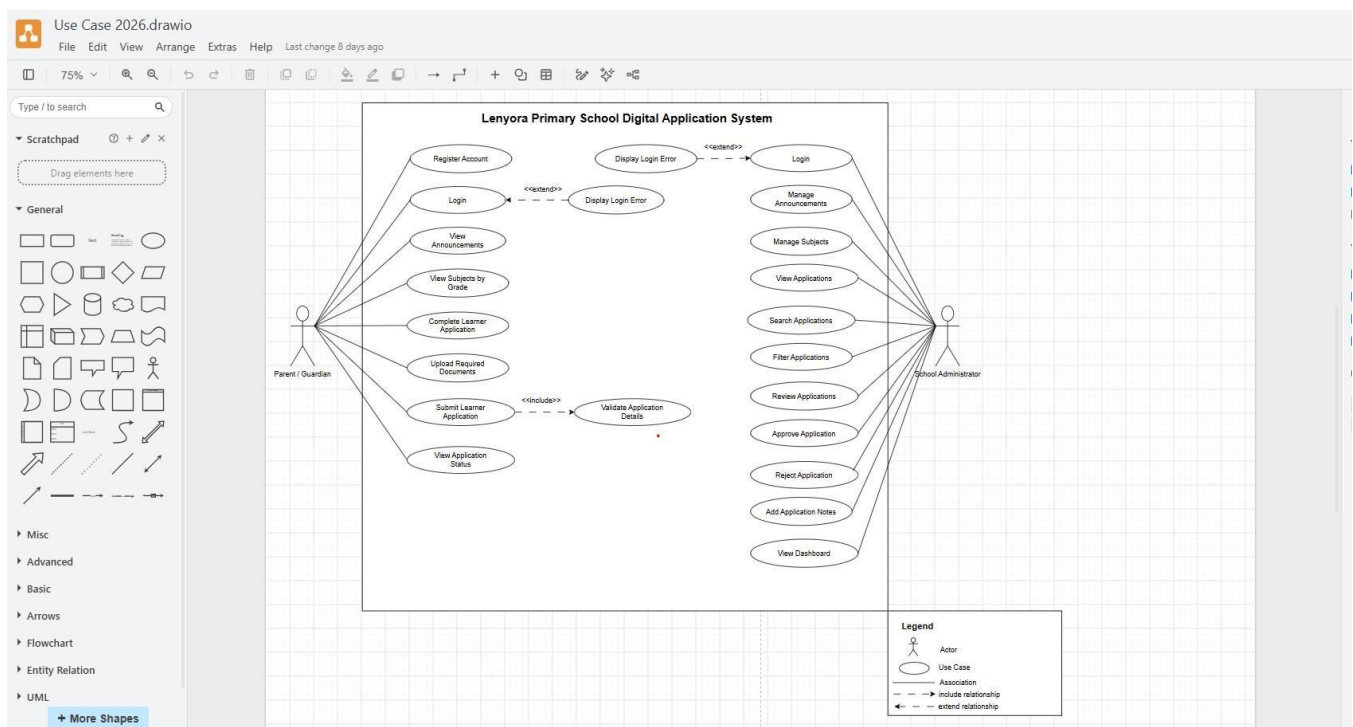
Supabase PostgreSQL	Shared users, announcements, subjects, learners, applications and document-reference data; access controlled with RLS.
Supabase Storage	Secure storage for required learner application documents such as IDs and birth certificates.
Supabase Realtime	Announcement/status update events where implemented.

2.1 Integration/Data Flow

- Parent/Guardian → Flutter: login details, learner information, application details and document-upload requests.
- Flutter → Supabase Auth/Storage: authentication and direct document upload operations.
- Flutter → ASP.NET Web API: REST/JSON application-related operations using authenticated requests.
- ASP.NET Web App → ASP.NET Web API → Supabase: administrator operations and database queries/updates.
- Shared Supabase data connects both applications: administrator-managed announcements/subjects can be displayed to parents, and parent-submitted applications can be reviewed by administrators.

3. UML Design

3.1 Actors and Use Cases

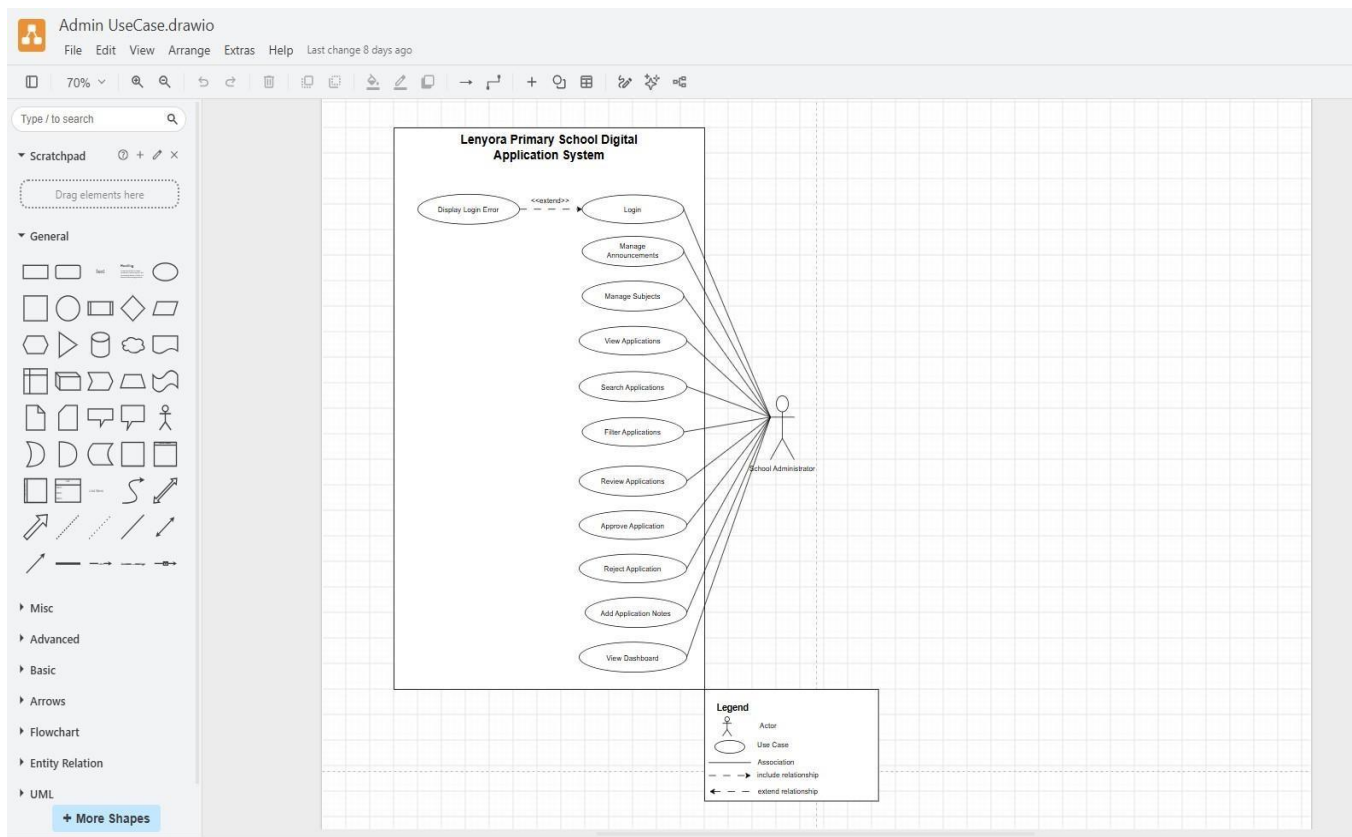


Combined Use Case Diagram

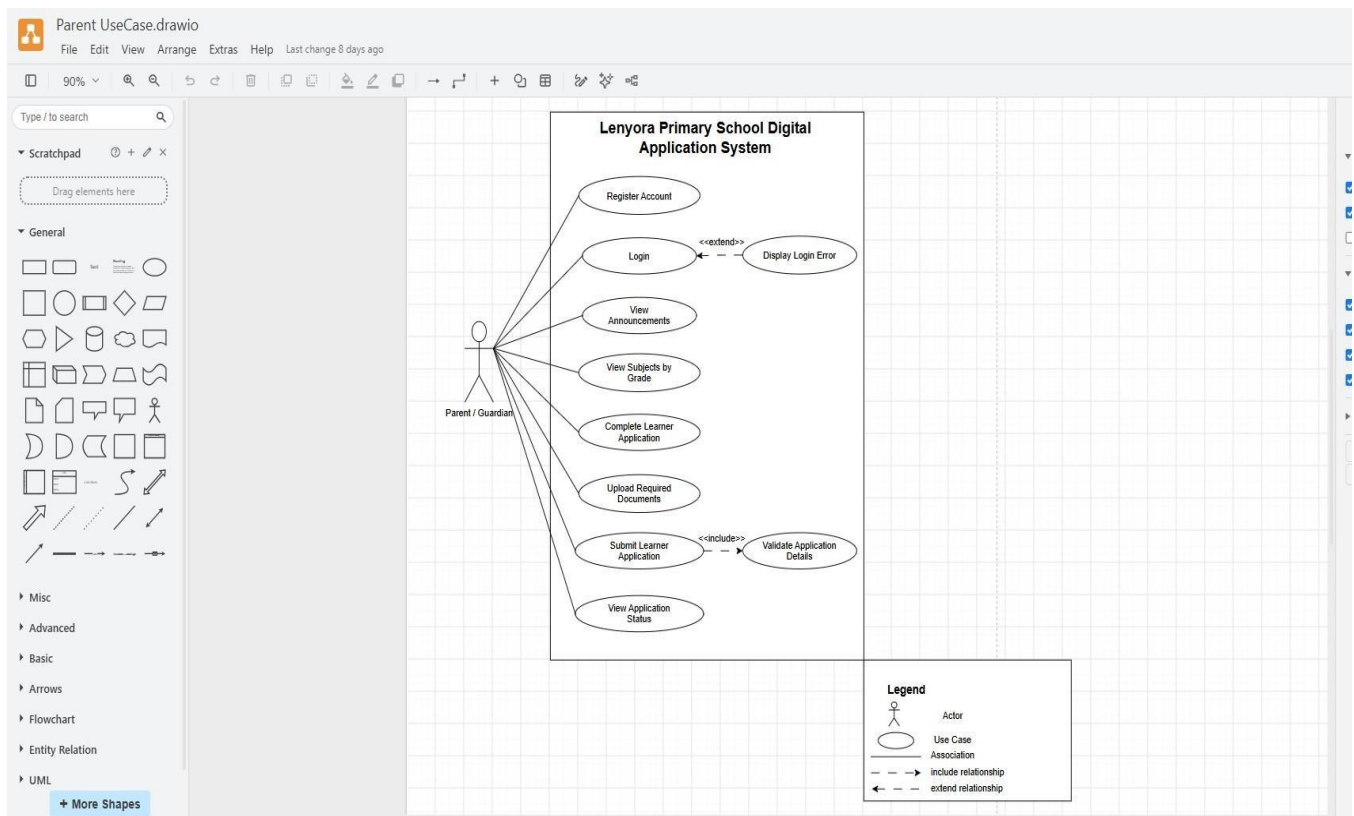
- Parent/Guardian: Register Account, Log In, View Announcements, View Subjects by Grade, Complete Learner Application, Upload Required Documents, Submit Learner Application and View Application Status.
- School Administrator: Log In, Manage Announcements, Manage Subjects, View Applications, Search Applications, Filter Applications, Review Application, Approve Application, Reject Application, Add Application Notes and View Dashboard.
- Guest: View General School Information, if retained as an approved requirement.

3.1.1 USE CASE DIAGRAMS

School Administrator role-specific use case screenshot.

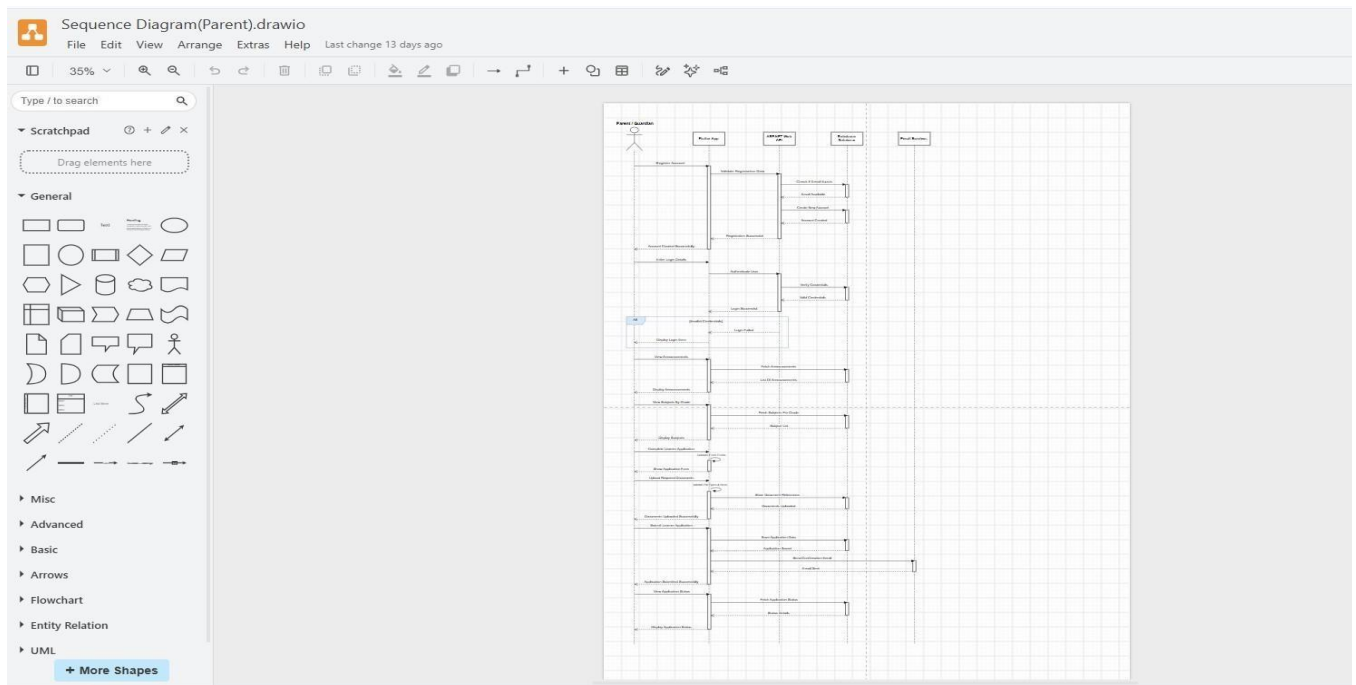


Parent/ Guardian role-specific use case screenshot.



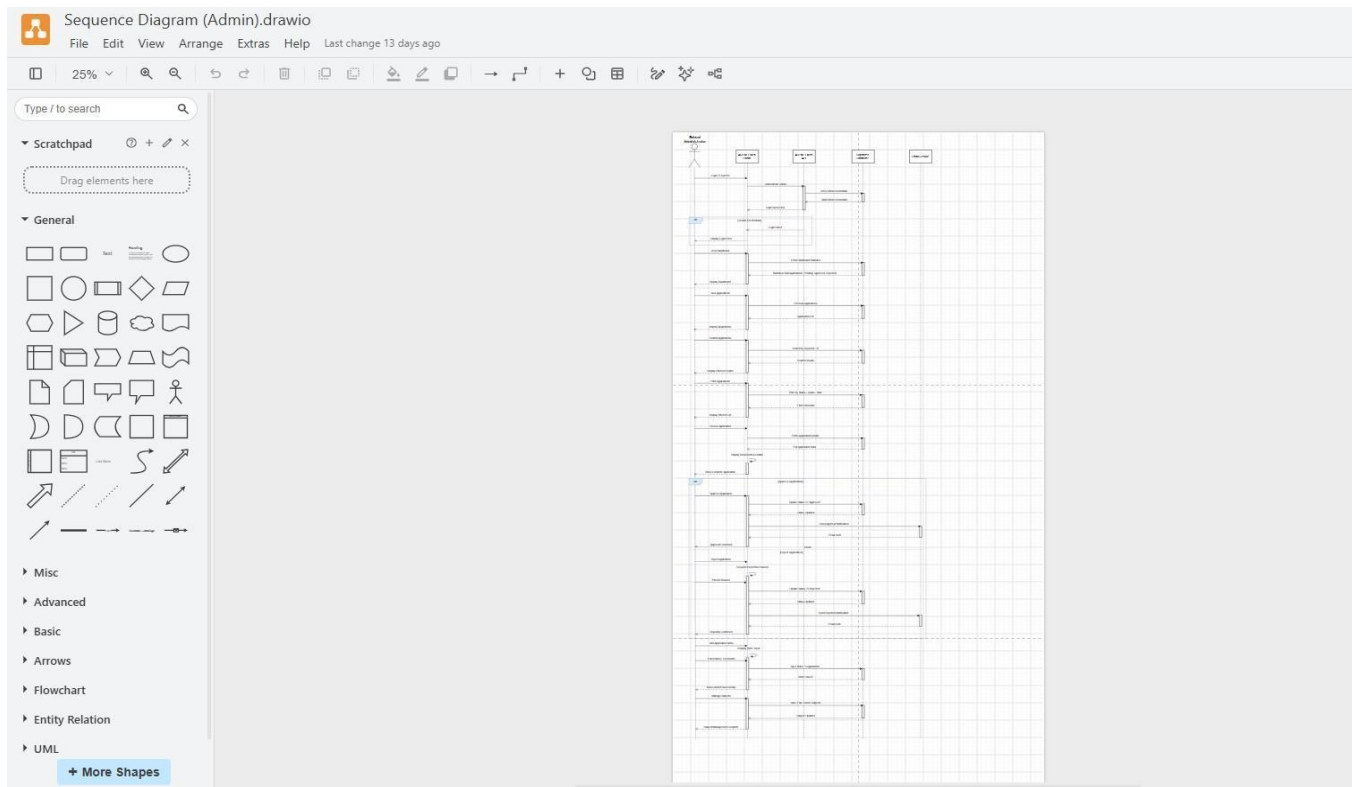
3.2 Sequence Diagram 1 — Flutter + Supabase

Approved-use-case alignment: Submit Learner Application. The parent submits through Flutter; required documents are uploaded to Supabase Storage; the application is sent through the ASP.NET Web API; the API validates and saves the application in Supabase PostgreSQL; and the resulting Pending status is returned to Flutter.



3.3 Sequence Diagram 2 — ASP.NET + Supabase

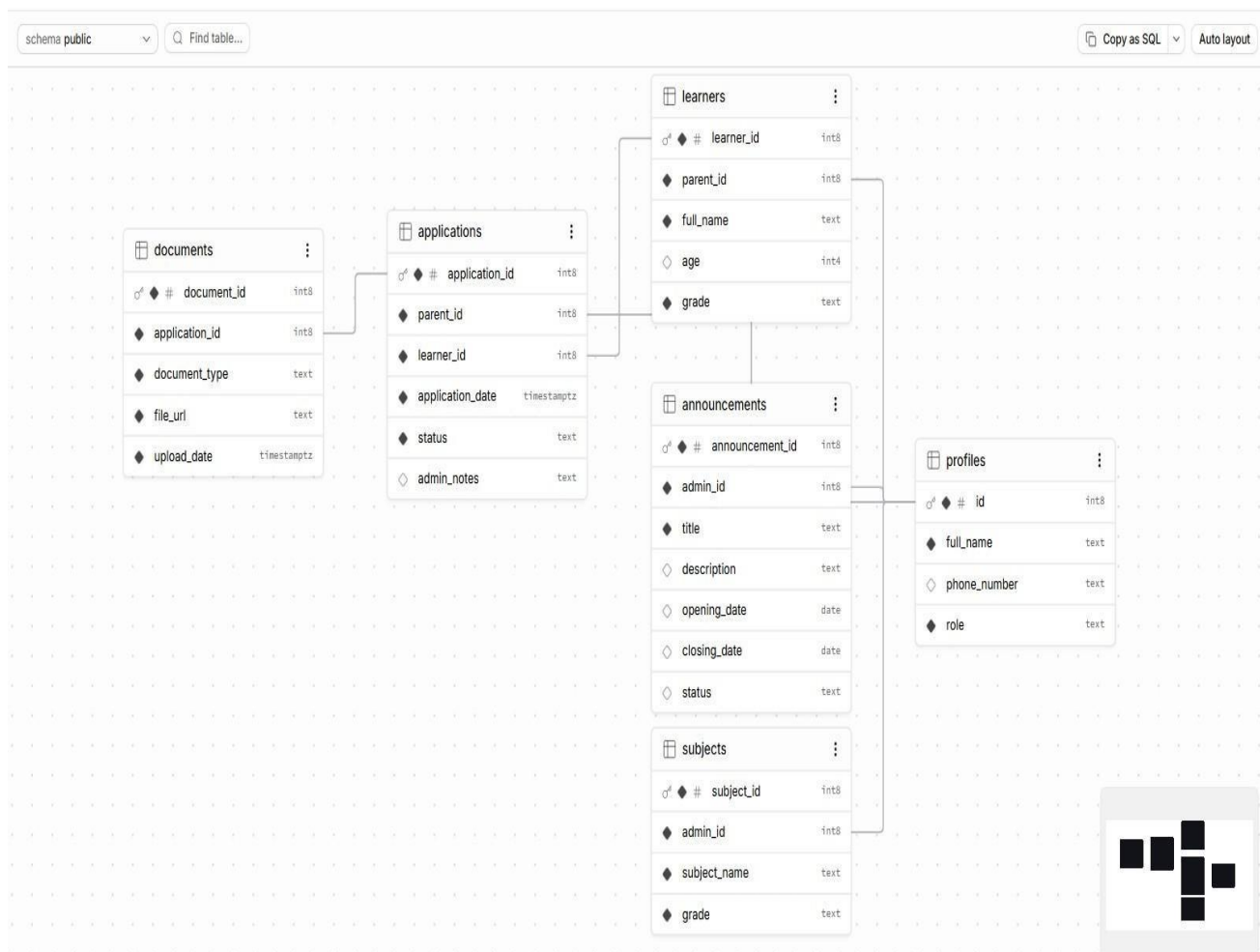
Approved-use-case alignment: Review Application, Approve Application and Reject Application. The administrator uses the ASP.NET web application; the web layer requests application data through the ASP.NET API; the API reads/writes Supabase PostgreSQL; and the decision is returned to the administrator interface.



4.Database Design — ERD

The current Supabase schema screenshot supplied for Phase 2 shows six main tables: profiles, announcements, subjects, learners, applications and documents. The visible fields and types are: profiles (id int8 PK, full_name text, phone_number text, role text); announcements (announcement_id int8 PK, admin_id int8 FK, title text, description text, opening_date date, closing_date date, status text); subjects (subject_id int8 PK, admin_id int8 FK, subject_name text, grade text); learners (learner_id int8 PK, parent_id int8 FK, full_name text, age int4, grade text); applications (application_id int8 PK, parent_id int8 FK, learner_id int8 FK, application_date timestampz, status text, admin_notes text); and documents (document_id int8 PK, application_id int8 FK, document_type text, file_url text, upload_date timestampz).

Supabase database schema screenshot (ERD)



4.1 Entities and Relationships

Relationship	Cardinality	Purpose	Used by
profiles → announcements	1:M	One administrator can create/manage many announcements	ASP.NET + Flutter
profiles → subjects	1:M	One administrator can manage many subject records	ASP.NET + Flutter
profiles → learners	1:M	One parent/guardian can have multiple learners	Flutter
profiles → applications	1:M	One parent/guardian can submit multiple applications	Flutter + ASP.NET
learners → applications	1:M	A learner can have application records	Flutter + ASP.NET

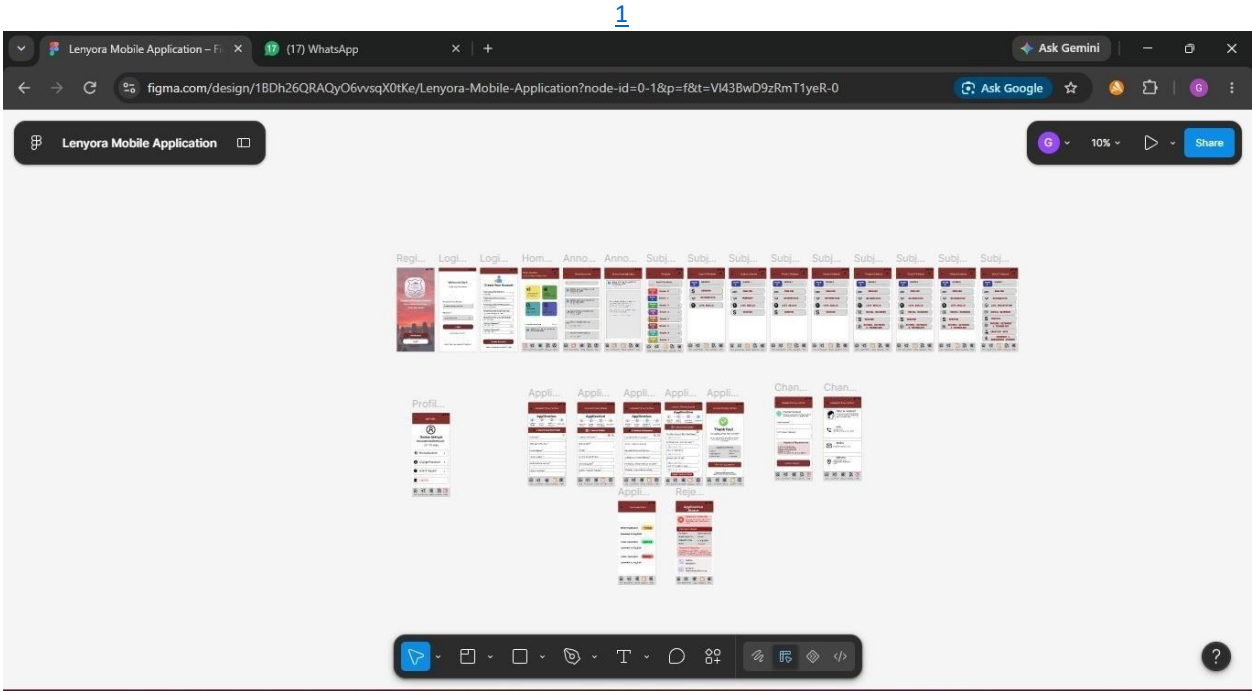
applications → documents	1:M	One application can have multiple uploaded documents	Flutter + ASP.NET
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5. User Interface Design

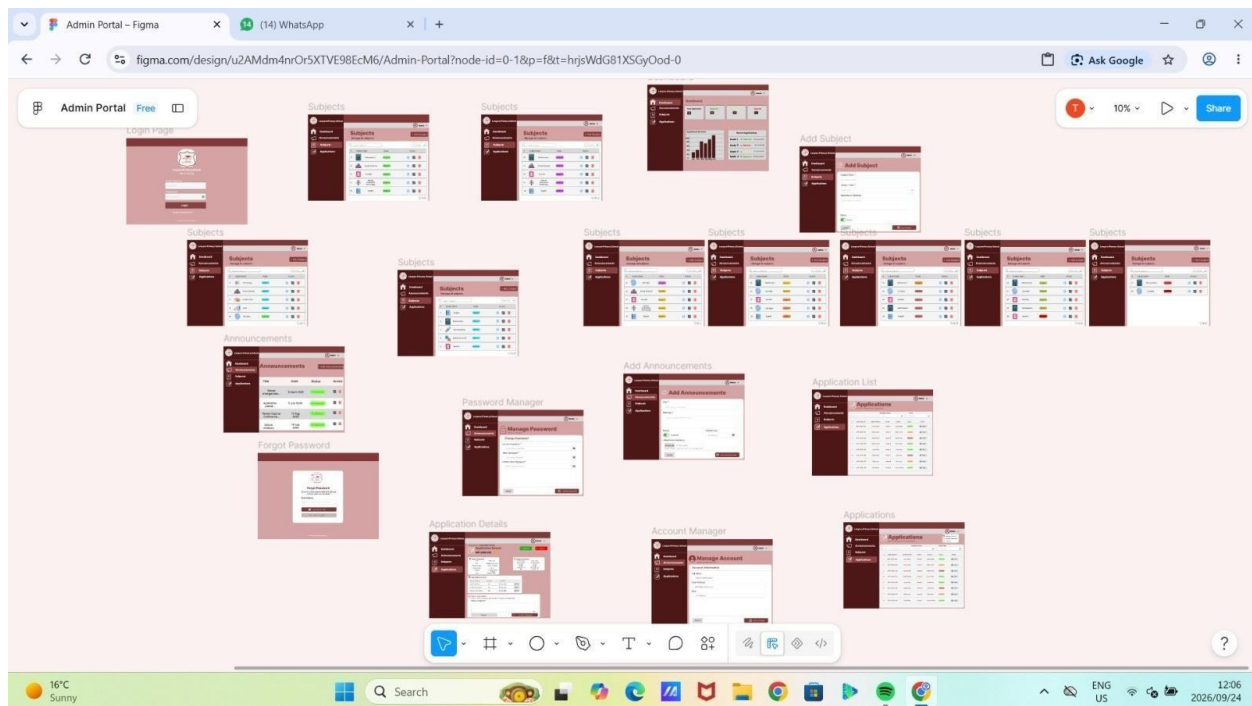
The group supplied two Figma sources. The mobile source covers the parent/guardian flow, while the admin source covers the school administrator portal.

Flutter / Lenyora Mobile Application Figma:

[https://www.figma.com/design/1BDh26QRAQyO6vvsqX0tKe/Lenyora-Mobile-Application?node-id=01&t=fSZuc7SQ6WHKLpsC-](https://www.figma.com/design/1BDh26QRAQyO6vvsqX0tKe/Lenyora-Mobile-Application?node-id=01&t=fSZuc7SQ6WHKLpsC-1&p=f&t=V43BwD9zRmT1yeR-0)



ASP.NET / Admin Portal Figma: <https://www.figma.com/design/u2AMdm4nrOr5XTVE98EcM6/Admin-Portal?node-id=0-1&p=f&t=I3ivMxHMap9dWZGT-0>



5.1 Flutter Mobile Screens

- Register Screen
- Login Screen
- Home Screen
- Announcements Screen
- Subject/Grade Screen
- Application Form Screen
- Application Status Tracking
- Profile Screen
- Grade Subjects screen

5.2 ASP.NET Administrator Screens

- Administrator Login
- Dashboard
- Manage Announcements
- Manage Subjects
- View/Search/Filter Applications
- Review Application
- Approve Application
- Reject Application

- Add Application Notes

6.Design Alignment Matrix

Requirement / Function	Use Case	Sequence	ERD elements	Interface	Evidence
Parent registration/login	Register Account / Log In	Authentication step	profiles	Register + Login	Figma mobile
View announcements	View Announcements	Flutter reads announcement data	announcements	Announcements	Figma mobile
View subjects by grade	View Subjects by Grade	Flutter reads subject data	subjects	Subjects/Grade	Figma mobile
Submit learner application	Complete / Upload / Submit Learner Application	Sequence 1	learners, applications, documents	Application Form	Figma mobile + draw.io
Track status	View Application Status	Sequence 1 response/status	applications	Status Tracking	Figma mobile
Manage announcements	Manage Announcements	Sequence 2/admin operations	announcements	Manage Announcements	Figma web
Manage subjects	Manage Subjects	Sequence 2/admin operations	subjects	Manage Subjects	Figma web
Review/search/filter	View/Search/Filter/ Review Applications	Sequence 2	applications, learners, documents	Application Management	Figma web
Approve/reject	Approve Application / Reject Application	Sequence 2	applications	Review/Decision	Figma web
Add notes	Add Application Notes	Sequence 2	applications.admin_n otes	Review/Decision	Figma web

7. Original Design Source Register

Diagrams.net source link 1:

https://app.diagrams.net/#G1ovZgl9FSmmSiOq3HWeynajoiPhARosHP#%7B%22pageId%22%3A%22FYLm_7UJ_IF0Y7jQHJy%22%7D

Diagrams.net source link 2:

<https://app.diagrams.net/#G1t5FyV4Oua4eC0xWsTkSk8FE98cL4BjyT#%7B%22pageId%22%3A%22GKN8Nx92uGMVM1phwASG%22%7D>

Diagrams.net source link 3:

https://app.diagrams.net/#G1ovZgl9FSmmSiOq3HWeynajoiPhARosHP#%7B%22pageId%22%3A%22FYLm_7UJ_IF0Y7jQHJy%22%7D

Diagrams.net source link 4:

<https://app.diagrams.net/#G1abTftLMlkalRIkt2om7Xn2U5tfQ7nPLc#%7B%22pageId%22%3A%22zkvKEWpfVYuaILGDTLmL%22%7D>

Diagrams.net source link 5:

https://app.diagrams.net/#G1fO4RMaF3fYIa0VDGGkbkInuRlzmzKIUB#%7B%22pageId%22%3A%22CRE5hrB6Mg_OrWMkIIfI%22%7D

Diagrams.net source link 6: https://app.diagrams.net/#G1ZM8cyl86FI_aNa_Y-fKMEN0wQg9pDjLa#%7B%22pageId%22%3A%22Nx74ppw-eD1Q8O4OuG4Q%22%7D

Figma — Flutter / Lenyora Mobile Application:

<https://www.figma.com/design/1BDh26QRAQyO6vvsgX0tKe/Lenyora-Mobile-Application?node-id=01&t=fSZuc7SQ6WHKLpsC-1>

Figma — ASP.NET Admin Portal: <https://www.figma.com/design/u2AMdm4nrOr5XTVE98EcM6/Admin-Portal?node-id=01&p=f&t=l3ivMxHMap9dWZGT-0>

GitHub repository: <https://github.com/hlonikayane/SCHOOL-DIGITAL-APPLICATION-SYSTEM>

Supabase project reference <https://tociqhbmenzsltuskaa.supabase.co>

8. Group, GitHub Project Evidence

GitHub repository: <https://github.com/hlonikayane/SCHOOL-DIGITAL-APPLICATION-SYSTEM>

Student no.	Member	GitHub identity supplied	Phase 2 responsibility to evidence
222002417	L Kayane	@hlonikayane	Flutter + Supabase integration
222020350	SM Nkosi	@222020350	SRS, documentation + coordination
222040802	TH Mosia	@TH Mosia	Architecture + UML + UI design
223028212	GD Sekhula	@gontseSE	Supabase database + security/verification
216009568	MN Matla	@fefematla60-png	ASP.NET administrator portal

9.Editable Design Source Register

Design	Editable source file	Tool
System Architecture	System Architecture.drawio	diagrams.net
Combined Use Case Diagram	Combined Use Case.drawio	diagrams.net
Sequence Diagram 1 — Parent	Sequence Diagram (Parent).drawio	diagrams.net
Sequence Diagram 2 — Administrator	Sequence Diagram (Admin).drawio	diagrams.net
ERD	Lenyora Database ERD	Supabase
Flutter Mobile UI	Lenyora Mobile Application (online Figma source)	Figma
ASP.NET Administrator UI	Admin Portal (online Figma source)	Figma

10. Required Online Links

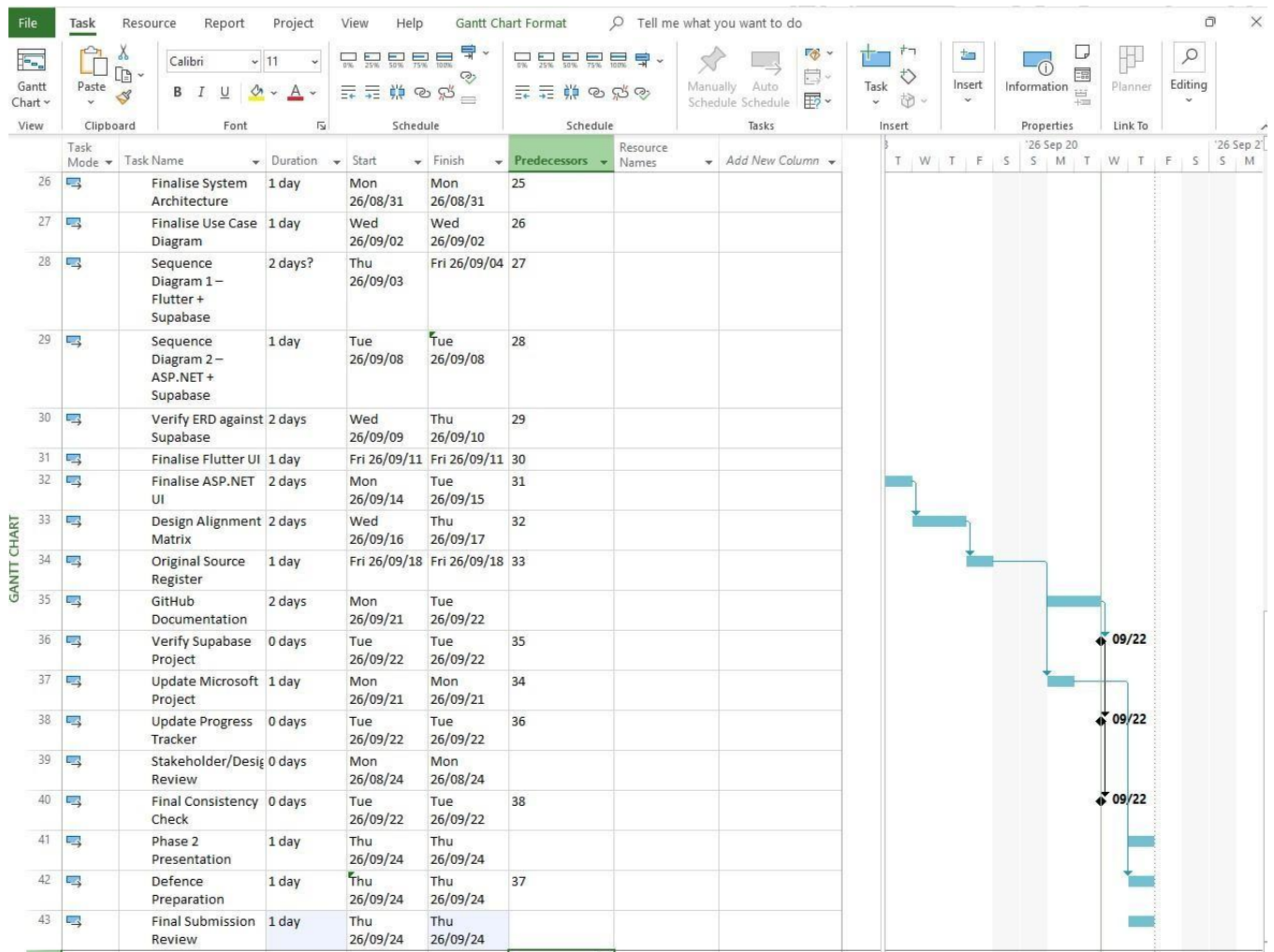
GitHub repository: [GitHub repository](#)

Supabase project reference: [Supabase project reference](#)

Flutter / Lenyora Mobile Application Figma: [Flutter / Lenyora Mobile Application Figma](#)

ASP.NET / Admin Portal Figma: [ASP.NET / Admin Portal Figma](#)

11.Microsoft Project Screenshot



12. Project Progress Tracker Evidence

Task ID	Reporting Phase	Activity	Task Description	Assigned Student	Start Date	Due Date	Status	% Complete	Evidence / Link	Verified by Group	Lecturer Comment	Progress Flag
24	U1-22	Unit 2 / Phase 1	Activity 2.2	Creating SWOT analysis	M.N. Matia	28/08/2026	31/08/2026	Completed	100%		Yes	Completed
25	U3-01	Phase 2	Activity 3.1	Finalise system architecture: components, responsibilities and data flows	T.H. Mosia	08/09/2026	10/09/2026	Completed	100%	Phase 2 architecture material / SRS Phase 2 section	Yes	Completed
26	U3-02	Phase 2	Activity 3.2	Finalise combined UML use case diagram with system boundary, actors and relationships	T.H. Mosia + All Members	08/09/2026	12/09/2026	Completed	100%	Phase 2 use-case material; final notation/consistency check required	Yes	Completed
27	U3-03	Phase 2	Activity 3.2	Finalise Sequence Diagram 1: Flutter + Supabase for learner application submission	L. Kayane + T.H. Mosia	09/09/2026	13/09/2026	Completed	100%	Phase 2 sequence material; verify messages against implemented flow	Yes	Completed
28	U3-04	Phase 2	Activity 3.2	Finalise Sequence Diagram 2: ASP.NET + Supabase for application review/decision	M.N. Matia + T.H. Mosia	09/09/2026	13/09/2026	Completed	100%	Phase 2 sequence material; verify API/database flow	Yes	Completed
29	U3-05	Phase 2	Activity 3.2	Verify ERD entities, attributes, PKs, FKs and cardinalities against the current Supabase database	G.D. Sekhula + L. Kayane	10/09/2026	14/09/2026	Completed	100%	Supabase project + ERD material; actual schema must be checked before submission	Yes	Completed
30	U3-06	Phase 2	Activity 3.3	Finalise Flutter UI screens and map each screen to approved use cases/requirements	L. Kayane + T.H. Mosia	10/09/2026	15/09/2026	Completed	100%	Figma mobile design; add requirement/use-case mapping	Yes	Completed
31	U3-07	Phase 2	Activity 3.3	Finalise ASP.NET administrator UI pages and map pages to admin use cases	M.N. Matia + T.H. Mosia	10/09/2026	15/09/2026	Completed	100%	Figma admin portal; add requirement/use-case mapping	Yes	Completed
32	U3-08	Phase 2	Activity 3.4	Complete design-alignment matrix: requirement → use case → sequence → ERD → interface	S.M. Nkosi + All Members	15/09/2026	17/09/2026	Completed	100%	Phase 2 SRS alignment table; final group verification required	Yes	Completed
33	U3-09	Phase 2	Activity 3.4	Complete original design source register with editable architecture/UML/ERD/UI links	S.M. Nkosi + All Members	15/09/2026	18/09/2026	Completed	100%	Needs real editable links/files and access checks	Yes	Completed
34	U3-10	Phase 2	Activity 3.4	Organise GitHub Phase 2 documentation, README, design artefacts and evidence	L. Kayane + S.M. Nkosi	16/09/2026	18/09/2026	Completed	100%	GitHub repository: https://github.com/hlonikayane/SCHOOL-DIGITAL-APPLICATION-SYSTEM	Yes	Completed
35	U3-11	Phase 2	Activity 3.4	Verify current Supabase project URL and add/confirm group members with appropriate access	L. Kayane + G.D. Sekhula	16/09/2026	18/09/2026	Completed	100%	Current project URL: https://venvjshduuvbfqqa.supabase.co	Yes	Completed
36	U3-12	Phase 2	Activity 3.4	Update editable Microsoft Project schedule with Phase 2 tasks, dates, dependencies, responsibilities and milestones	T.H. Mosia + M.N. Matia	17/09/2026	19/09/2026	Completed	100%	Updated .mpp and screenshot required	Yes	Completed
37	U3-13	Phase 2	Activity 3.4	Update Phase 2 progress tracker and individual contribution evidence	S.M. Nkosi + All Members	17/09/2026	22/09/2026	Completed	100%	This updated tracker; individual evidence still requires group verification	Yes	Completed
38	U3-14	Phase 2	Activity 3.4	Conduct stakeholder/design review and record feedback or confirmation	M.N. Matia + All Members	18/09/2026	20/09/2026	Completed	100%	Meeting record/feedback evidence required if a review is conducted	Yes	Completed
39	U3-15	Phase 2	Activity 3.4	Perform final consistency and requirements traceability check across SRS, UML, ERD, UI and architecture	G.D. Sekhula + All Members	19/09/2026	20/09/2026	Completed	100%	Check terminology, fields, flows and approved requirements	Yes	Completed
40	U3-16	Phase 2	Activity 3.5	Prepare Phase 2 presentation covering architecture, UML, sequences, ERD, UI, alignment and progress	All Members	19/09/2026	21/09/2026	Completed	100%	Phase 2 presentation deck; insert current progress evidence	Yes	Completed
41	U3-17	Phase 2	Activity 3.5	Prepare individual defence questions and ensure each member can explain their contribution	All Members	20/09/2026	22/09/2026	Completed	100%	Individual defence preparation	Yes	Completed
42	U3-18	Phase 2	Activity 3.5	Final Phase 2 submission package review and submission readiness check	All Members	22/09/2026	22/09/2026	Completed	100%	Final checklist: SRS, diagrams, source links, .mpp, tracker, GitHub evidence	Yes	Completed